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24432028
COHORT B

MariaDB [(none)]> create database Projects;
Query OK, 1 row affected (0.010 sec)

MariaDB [(none)]> use Projects;
Database changed

QUESTION 1

MariaDB [Projects]> CREATE TABLE Employee
-> (
->
-> empNo CHAR(5) PRIMARY KEY,
->
-> fName VARCHAR(30) NOT NULL,
->
-> lName VARCHAR(30) NOT NULL,
->
-> address VARCHAR(100),
->
-> DOB DATE,
->
-> sex CHAR(1)
->
-> CHECK (sex IN ('M','F')),
->
-> position VARCHAR(25)
->
-> CHECK (position IN
->
-> ('Manager', 'Team Leader', 'Analyst', 'Software Developer')),
->
-> deptNo CHAR(5)
->
->);
Query OK, 0 rows affected (0.058 sec)

MariaDB [Projects]> CREATE TABLE Department
-> (
->
-> deptNo CHAR(5) PRIMARY KEY,
->
-> deptName VARCHAR(50) NOT NULL,
->
-> mgrEmpNo CHAR(5)

```
->  
-> )  
-> ;
```

Query OK, 0 rows affected (0.015 sec)

MariaDB [Projects]> CREATE TABLE Project

```
-> (  
->  
-> projNo CHAR(5) PRIMARY KEY,  
->  
-> projName VARCHAR(50) NOT NULL,  
->  
-> deptNo CHAR(5),  
->  
->  
->  
-> FOREIGN KEY (deptNo)  
->  
-> REFERENCES Department(deptNo)  
->  
-> );
```

Query OK, 0 rows affected (0.020 sec)

MariaDB [Projects]> CREATE TABLE WorksOn

```
-> (  
->  
-> empNo CHAR(5),  
->  
-> projNo CHAR(5),  
->  
-> dateWorked DATE,  
->  
-> hoursWorked INT  
->  
-> CHECK (hoursWorked BETWEEN 0 AND 40),  
->  
-> PRIMARY KEY (empNo, projNo, dateWorked),  
->  
-> FOREIGN KEY (empNo)  
->  
-> REFERENCES Employee(empNo),  
->  
-> FOREIGN KEY (projNo)  
->  
-> REFERENCES Project(projNo)  
->  
-> );
```

Query OK, 0 rows affected (0.018 sec)

MariaDB [Projects]> ALTER TABLE Employee

```
-> ADD CONSTRAINT FK_Employee_Department
-> FOREIGN KEY (deptNo)
-> REFERENCES Department(deptNo)
-> ;
```

Query OK, 0 rows affected (0.041 sec)
Records: 0 Duplicates: 0 Warnings: 0

```
MariaDB [Projects]> ALTER TABLE Department
-> ADD CONSTRAINT FK_Department_Manager
-> FOREIGN KEY (mgrEmpNo)
-> REFERENCES Employee(empNo);
```

Query OK, 0 rows affected (0.028 sec)
Records: 0 Duplicates: 0 Warnings: 0

```
MariaDB [Projects]> INSERT INTO Department (deptNo, deptName, mgrEmpNo) VALUES
```

```
-> ('D01','Human Resources',NULL),
-> ('D02','Information Technology',NULL),
-> ('D03','Finance',NULL),
-> ('D04','Marketing',NULL),
-> ('D05','Operations',NULL),
-> ('D06','Research',NULL),
-> ('D07','Sales',NULL),
-> ('D08','Customer Service',NULL),
-> ('D09','Engineering',NULL),
-> ('D10','Procurement',NULL),
-> ('D11','Legal',NULL),
-> ('D12','Administration',NULL);
```

Query OK, 12 rows affected (0.015 sec)
Records: 12 Duplicates: 0 Warnings: 0

```
MariaDB [Projects]> INSERT INTO Employee
```

```
-> (empNo,fName,lName,address,DOB,sex,position,deptNo)
-> VALUES
-> ('E01','Ama','Mensah','Accra','1990-03-15','F','Manager','D01'),
-> ('E02','Kwame','Asante','Kumasi','1992-07-20','M','Manager','D02'),
-> ('E03','Akosua','Boateng','Cape Coast','1995-01-12','F','Team Leader','D03'),
-> ('E04','Kojo','Owusu','Takoradi','1993-08-11','M','Analyst','D04'),
-> ('E05','Linda','Agyeman','Tema','1994-05-17','F','Software Developer','D05'),
-> ('E06','Yaw','Ofori','Accra','1991-10-30','M','Manager','D06'),
-> ('E07','Abena','Adjei','Sunyani','1996-02-22','F','Analyst','D07'),
-> ('E08','Kofi','Appiah','Ho','1992-11-08','M','Software Developer','D08'),
-> ('E09','Grace','Arthur','Tamale','1997-09-14','F','Team Leader','D09'),
-> ('E10','Daniel','Mensimah','Koforidua','1990-04-09','M','Manager','D10'),
-> ('E11','Esi','Quaye','Kasoa','1994-06-01','F','Software Developer','D11'),
-> ('E12','Michael','Addo','Accra','1993-12-18','M','Analyst','D12');
```

Query OK, 12 rows affected (0.010 sec)
Records: 12 Duplicates: 0 Warnings: 0

```
MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E01' WHERE deptNo='D01';
```

Query OK, 1 row affected (0.007 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E02' WHERE deptNo='D02';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E03' WHERE deptNo='D03';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E04' WHERE deptNo='D04';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E05' WHERE deptNo='D05';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E06' WHERE deptNo='D06';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E07' WHERE deptNo='D07';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E08' WHERE deptNo='D08';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E09' WHERE deptNo='D09';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E10' WHERE deptNo='D10';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E11' WHERE deptNo='D11';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E12' WHERE deptNo='D12';
Query OK, 1 row affected (0.007 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [Projects]> INSERT INTO Project
-> (projNo,projName,deptNo)
-> VALUES

```

-> ('P01','Payroll System','D01'),
-> ('P02','Student Portal','D02'),
-> ('P03','Accounting App','D03'),
-> ('P04','Marketing Website','D04'),
-> ('P05','Inventory System','D05'),
-> ('P06','AI Research','D06'),
-> ('P07','Sales Dashboard','D07'),
-> ('P08','Help Desk','D08'),
-> ('P09','Mobile Banking','D09'),
-> ('P10','Supplier Tracker','D10'),
-> ('P11','Legal Archive','D11'),
-> ('P12','Office Automation','D12');

```

Query OK, 12 rows affected (0.007 sec)

Records: 12 Duplicates: 0 Warnings: 0

MariaDB [Projects]> INSERT INTO WorksOn

```

-> (empNo,projNo,dateWorked,hoursWorked)
-> VALUES
-> ('E01','P01','2026-01-10',35),
-> ('E02','P02','2026-01-10',32),
-> ('E03','P03','2026-01-11',28),
-> ('E04','P04','2026-01-11',20),
-> ('E05','P05','2026-01-12',38),
-> ('E06','P06','2026-01-12',30),
-> ('E07','P07','2026-01-13',25),
-> ('E08','P08','2026-01-13',22),
-> ('E09','P09','2026-01-14',40),
-> ('E10','P10','2026-01-14',34),
-> ('E11','P11','2026-01-15',36),
-> ('E12','P12','2026-01-15',27);

```

Query OK, 12 rows affected (0.009 sec)

Records: 12 Duplicates: 0 Warnings: 0

QUESTION 2

MariaDB [Projects]> CREATE VIEW FemaleManagedProjects as

```

->
-> SELECT
->
-> p.projNo,
->
-> p.projName,
->
-> e.empNo,
->
-> e.fName,
->
-> e.lName
->
-> FROM Project p

```

```

->
-> JOIN Department d
->
-> ON p.deptNo = d.deptNo
->
-> JOIN Employee e
->
-> ON d.mgrEmpNo = e.empNo
->
-> WHERE e.sex = 'F'
->
-> ORDER BY p.projNo;
Query OK, 0 rows affected (0.004 sec)

```

MariaDB [Projects]> select * from femalemanagedprojects;

projNo	projName	deptName	empNo	fName	lName
P01	Payroll System	Human Resources	E01	Ama	Mensah
P03	Accounting App	Finance	E03	Akosua	Boateng
P05	Inventory System	Operations	E05	Linda	Agyeman
P07	Sales Dashboard	Sales	E07	Abena	Adjei
P09	Mobile Banking	Engineering	E09	Grace	Arthur
P11	Legal Archive	Legal	E11	Esi	Quaye

6 rows in set (0.008 sec)

QUESTION 3

MariaDB [Projects]> CREATE VIEW EmployeeProjectHours as

```

->
-> SELECT
->
-> e.empNo,
->
-> e.fName,
->
-> e.lName,
->
-> p.projName,
->
-> w.hoursWorked
->
-> FROM Employee e
->
-> JOIN WorksOn w
->
-> ON e.empNo = w.empNo
->
-> JOIN Project p

```

```

->
-> ON w.projNo = p.projNo
->
-> ;
Query OK, 0 rows affected (0.014 sec)

```

MariaDB [Projects]> select * from employeeprojecthours;

empNo	fName	lName	projName	hoursWorked
E01	Ama	Mensah	Payroll System	35
E02	Kwame	Asante	Student Portal	32
E03	Akosua	Boateng	Accounting App	28
E04	Kojo	Owusu	Marketing Website	20
E05	Linda	Agyeman	Inventory System	38
E06	Yaw	Ofori	AI Research	30
E07	Abena	Adjei	Sales Dashboard	25
E08	Kofi	Appiah	Help Desk	22
E09	Grace	Arthur	Mobile Banking	40
E10	Daniel	Mensimah	Supplier Tracker	34
E11	Esi	Quaye	Legal Archive	36
E12	Michael	Addo	Office Automation	27

12 rows in set (0.016 sec)

QUESTION 4

MariaDB [Projects]> CREATE VIEW EmpProject(empNo, projNo, totalHours) as

```

->
-> SELECT
->
-> w.empNo,
->
-> w.projNo,
->
-> SUM(hoursWorked)
->
-> FROM Employee e,
->
-> Project p,
->
-> WorksOn w
->
-> WHERE e.empNo = w.empNo
->
-> AND p.projNo = w.projNo
->
-> GROUP BY w.empNo, w.projNo
->
-> ;

```

Query OK, 0 rows affected (0.012 sec)

(a). MariaDB [Projects]> select * from empproject;

empNo	projNo	totalHours
E01	P01	35
E02	P02	32
E03	P03	28
E04	P04	20
E05	P05	38
E06	P06	30
E07	P07	25
E08	P08	22
E09	P09	40
E10	P10	34
E11	P11	36
E12	P12	27

12 rows in set (0.014 sec)

(b). MariaDB [Projects]> SELECT projNo

-> FROM EmpProject

-> WHERE projNo = 'P01';

projNo
P01

1 row in set (0.008 sec)

(c). MariaDB [Projects]> SELECT COUNT(projNo)

-> FROM EmpProject

-> WHERE empNo = 'E01';

COUNT(projNo)
1

1 row in set (0.001 sec)

(d). MariaDB [Projects]> SELECT

-> empNo,

-> SUM(totalHours) AS totalHours

-> FROM EmpProject

-> GROUP BY empNo;

empNo	totalHours
E01	35

E01	35
E02	32
E03	28
E04	20
E05	38
E06	30
E07	25
E08	22
E09	40
E10	34
E11	36
E12	27

+-----+

12 rows in set (0.002 sec)

QUESTION 4

```
MariaDB [Projects]> CREATE VIEW ExpensiveParts(partNo) as
->
-> SELECT DISTINCT partNo
->
-> FROM Part
->
-> WHERE partCost > 1000;
```

The materialized view stores the results of the query instead of calculating them every time. Whenever the Part table changes, the materialized view should also be updated. If a new part with a cost greater than 1000 is added, we include its partNo in the view. If an expensive part is deleted or its cost drops below 1000, remove it from the view if it no longer meets the condition.

The base table does not need to be accessed when adding a new expensive part because the new row contains all the required information. Also, the base table must be accessed when deleting a part or reducing its cost, to check whether another record for the same part still has a cost greater than 1000.